MOBILE COMPUTING

ASSIGNMENT 1 – CO1

DESIGN, ANALYSE AND OPTIMISE A HEXAGONAL CELLULAR NETWORK

Student details	**Information**
Name	K. Goutham
Register Number	192311169
Programme / Section	B.E -CSE
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Abstract

This report develops a frequency-reuse and capacity plan for a 12 km² university township using a fixed allocation of 336 duplex traffic channels. It compares three distinct approaches: (i) fixed hexagonal cluster reuse, (ii) sectorised reuse, and (iii) hierarchical macro–micro deployment with selective cell splitting and dynamic channel borrowing. The design uses a seven-cell reuse cluster, a 12 dB minimum digital-style SIR planning threshold, and a 2% busy-hour blocking target. Hexagonal geometry gives a reuse distance of $D = R\sqrt{(3N)}$, cell area of $(3\sqrt{3}/2)R^2$, and 48 channels per cell for N=7. Erlang-B analysis gives approximately 38.39 Erlangs per cell at 2% blocking. A 19-cell initial layout therefore carries about 729 Erlangs, while uniform two-dimensional cell splitting by a factor of four per round gives 76 cells and approximately 2,917 Erlangs, sufficient for the 2,500-Erlang 100,000-subscriber forecast under the stated uniform-load approximation.

The event challenge adds 1,250 Erlangs to a 1 km² stadium zone for four hours. The report finds that the normal network cannot carry this concentrated surge without local augmentation. A temporary microcell/COW overlay, directional antennas, power control and pre-planned channel coordination are recommended rather than permanently shrinking every cell in the township. The report includes reproducible Python code; measured outputs should be generated by running that code in the stated environment.

Bloom's Taxonomy alignment

Bloom level	What this report demonstrates	Tasks
Remember / Understand	Define cellular reuse, cluster size, SIR, Erlang traffic, access and control channels.	1–3
Apply	Use hexagonal geometry, Erlang-B, recurrence relations and Python algorithms.	3, 6–9
Analyse	Compare interference, best/average/worst cases, scaling and theory–simulation gaps.	4, 5, 10
Evaluate	Judge strategy trade-offs, blocking, costs and event-day alternatives.	10–12
Create	Construct a multi-strategy plan, simulator workflow and final adaptive network design.	1, 8, 11, 12

Interpretation: The higher-weight tasks require analysis, evaluation and creation—not only definitions. Accordingly, each major result is derived, interpreted in the township context, and linked to a design decision.

Planning assumptions and selected parameters

Parameter	Adopted value / assumption	Reason
Township area	12 km²	Given scenario.
Available duplex traffic channels, S	336	Given, after control-channel overhead.
Cluster size, N	7	Conventional reuse choice; balances capacity and interference.
Channels per cell, S/N	48	336/7 exactly.

SIR threshold	12 dB	Digital-style minimum from the assignment range; conservative enough for reuse while not assuming analogue 18 dB.
Path-loss exponent, n	3.0 typical; 2.5 open / 4 dense sensitivity	Campus has open spaces and built-up blocks; n=3 is a planning midpoint, not a universal measured value.
Blocking target	≤2%	Given.
Busy-hour traffic / subscriber	0.025 E	Given.
Cell layout approximation	19 initial hexagonal cells; boundary cells treated by area-equivalent planning	Simplifies a finite township boundary; final site survey must refine it.
Traffic distribution	Uniform for base calculations; stadium surge treated separately	Explicitly exposes the failure of a uniform-demand assumption during events.

Important modelling note: A real network uses integer channel plans, control-channel reservations, guard bands, antenna patterns, terrain data and measured traffic maps. The 336-channel pool is treated as traffic-channel equivalents because the scenario says control overhead has already been removed.

1. Task 1 – Design alternative frequency-planning strategies (5 marks)

1.1 Strategy A: Fixed cluster frequency reuse

Basic idea: divide the available spectrum into N disjoint channel groups and repeat the groups in a regular hexagonal pattern. With N=7, each cell receives 48 duplex traffic channels. The same group is reused only at cells separated by the reuse distance $D=R\sqrt{(3N)}$. This is spatial reuse: geographic separation permits the same frequencies to be used again.

Assumptions: regular hexagonal cells; approximately equal cell radii; uniform channel demand; first-tier co-channel interference dominates; base-station powers and antenna patterns are controlled; a seven-cell cluster can be tiled over the planning region.

Access-channel interpretation: In an FDMA/TDMA implementation, each cell receives 48 duplex channel equivalents, and a traffic channel is allocated to a call. A control channel carries access requests, paging and system information and is not counted again in the 336 traffic-channel pool. In an OFDMA system, the same planning principle is applied to resource blocks/subcarriers and time slots rather than indivisible voice channels. CDMA shares a wider band by code separation but still needs reuse/interference control.

Iterative procedure: enumerate valid hexagonal cluster sizes $N=i^2+ij+j^2$; compute channels per cell, reuse ratio and estimated SIR; reject candidates that fail the SIR threshold; choose the smallest acceptable N that meets the interference constraint.

$$\mathbf{N = i^2 + ij + j^2 \quad ; \quad Q = D/R = \sqrt{(3N)}}$$

Pseudocode – fixed reuse

```
INPUT: S=336, township area A, candidate (i,j), path-loss n, SIR_min
FOR each integer pair (i,j):
   N = i*i + i*j + j*j
   channels_per_cell = floor(S/N)
```

```
  Q = sqrt(3*N)
  SIR_dB = 10*log10(( (sqrt(3*N))**n ) / 6)
  IF channels_per_cell > 0 AND SIR_dB >= SIR_min:
     estimate cell count from A and chosen R
     calculate Erlang-B capacity at 2% blocking
     retain candidate and its cost/capacity
SELECT feasible candidate with best cost–capacity trade-off
```

Preprocessing and cost bearer: the operator funds drive/walk tests, clutter and terrain maps, traffic heat maps, candidate-site surveys, frequency planning and optimisation. The university may provide site access and building information; it should not be assumed to bear radio-planning costs.

Why investigate: it is the simplest auditable baseline and gives a direct relationship between cluster size, channels per cell and co-channel interference.

1.2 Strategy B: Sectorised frequency reuse

Basic idea: retain a reuse cluster but divide a cell into directional sectors using 120° (three-sector) or 60° (six-sector) antennas. The antenna pattern suppresses signals arriving from directions outside the serving sector. This is spatial filtering: the number and strength of effective first-tier interferers are reduced.

Assumptions: sector antennas are correctly oriented; downtilt, azimuth and sidelobes are calibrated; sector boundaries are mapped; traffic can be distributed among sectors. Sectoring does not create additional spectrum. If the 48 channels are split into three isolated groups of 16, trunking efficiency decreases; if channels are pooled across sectors with coordinated scheduling, the trunking penalty can be reduced but coordination becomes more complex.

Access sharing: control channels identify the serving cell/sector; traffic channels or OFDMA resource blocks are scheduled within the sector. Handoffs may occur at sector boundaries even when the mobile remains inside the same physical site.

Pseudocode – sectorised reuse

```
INPUT: hex grid, N=7, 336 channels, sector count K=3
FOR each cell:
   create K directional sectors with measured antenna patterns
   assign/coordinate the cell's 48 channel-equivalents across sectors
   FOR each candidate mobile position:
      calculate desired received power using serving-sector gain
      calculate interference using each interferer's sector gain
      compute SIR and outage
   estimate sector traffic and Erlang-B blocking
IF any sector exceeds its blocking or SIR target:
   rebalance resources / adjust azimuth, tilt or sector boundaries
OUTPUT: sector SIR, blocking, handoffs and additional site/antenna cost
```

Preprocessing and cost bearer: the operator bears antenna-pattern measurements, sector azimuth planning, site visits, drive tests, installation and optimisation. The additional cost is justified where traffic is directional or concentrated, such as long academic corridors, roads and stadium stands.

Why investigate: sectoring may improve SIR without increasing licensed spectrum, but it must be evaluated against channel fragmentation, sector imbalance and added handoffs.

1.3 Strategy C: Hierarchical macro–micro network with selective splitting and borrowing

Basic idea: retain macro cells for broad coverage and mobility, then add microcells/picocells only in overloaded locations such as hostels, lecture halls, the arterial road and stadium. A controller may temporarily borrow idle channels from neighbouring cells where interference constraints permit. Repeated splitting is a recursive/hierarchical procedure: an overloaded parent cell is replaced by smaller child cells until its traffic is supportable.

Assumptions: demand is spatially non-uniform; candidate micro-sites and backhaul exist; neighbouring cells can coordinate borrowing; transmit power is reduced as cells shrink; handoff thresholds and neighbour lists are maintained. Borrowing is permitted only if the donor cell remains below its own blocking limit and the reuse/interference plan remains valid.

Pseudocode – recursive cell splitting and channel borrowing

```
PROCEDURE PLAN(cell, offered_traffic):
  IF ErlangB_blocking(cell.channels, offered_traffic) <= 0.02:
    RETURN cell
  ELSE:
    split cell into four child cells of radius R/2
    reduce child transmit power according to radius/path-loss model
    assign reuse-compatible channel groups to children
    FOR each child:
      PLAN(child, estimated_child_traffic)
    END FOR
    coordinate temporary borrowing from eligible neighbours
    RETURN child-cell layout
END PROCEDURE
```

Preprocessing and cost bearer: operator-funded granular traffic measurements, indoor/outdoor propagation surveys, backhaul checks, landlord permissions, small-cell installation, power/network integration and handoff testing. Borrowing is mainly a software/coordination cost; microcells and COWs create capital, rental and backhaul costs.

Why investigate: it scales spatially with demand and avoids permanently overbuilding low-traffic areas. Its disadvantages are higher site count, more handoffs, interference coordination and operational complexity.

1.4 Strategy comparison

Strategy	Main design principle	Main advantage	Main limitation
A. Fixed reuse	Spatial reuse	Simple, predictable, low operational complexity	Capacity per cell falls as N rises; cannot follow local hot spots.
B. Sectorised reuse	Spatial filtering	Improves SIR and directional capacity without more spectrum	Sector imbalance, handoffs and possible trunking loss.

C. Macro–micro / borrowing	Hierarchical reuse + trunking	Adds capacity where/when needed; supports growth and event overlays	More sites, coordination, backhaul and handoff complexity.

2. Task 2 – Identify the governing design parameter (2 marks)

Strategy	Governing parameter	Why it governs the design	Growth response
Fixed reuse	Cluster size N (equivalently $Q=\sqrt{3N}$)	N sets channels/cell S/N and reuse distance; larger N improves SIR but reduces per-cell channel share.	Keep N at the smallest value satisfying SIR; add cells/reduce R as traffic grows.
Sectorised reuse	Effective first-tier interferer count and sector antenna gain	SIR depends on both number and angular attenuation of co-channel interferers.	Optimise antenna pattern/sector load; avoid sector overload as users rise.
Hierarchical microcells	Cell radius R and number of active microcells	Smaller R increases site density and spatial reuse; it also increases handoff rate.	Split only overloaded cells; add temporary capacity for short surges.

The three parameters are coupled rather than independent. For example, reducing R does not change the ideal reuse ratio Q for a fixed N, but it increases the number of cells covering the township and therefore the total number of repeated channel groups.

3. Task 3 – Derive capacity and interference functions mathematically (6 marks)

3.1 Hexagonal geometry and number of cells

Let R be the centre-to-vertex cell radius (equal to the side length of a regular hexagon). A hexagon is composed of six equilateral triangles of side R. One triangle has area $(\sqrt{3}/4)R^2$; hence:

$$\mathbf{A_cell = 6 \times (\sqrt{3}/4)R^2 = (3\sqrt{3}/2)R^2 \approx 2.598R^2}$$

For a cluster of N cells, the cluster area is N A_cell. The number of cells needed for township area A_T is therefore:

$$\mathbf{M = ceil[A_T / A_cell] = ceil[12 / (2.598R^2)]}$$

Choosing $M_0=19$ for the initial area-equivalent layout gives $R_0 = \sqrt{(12/(19\times2.598))} = 0.493$ km and A_cell = 0.632 km². The estimated 19 cells cover 12 km² after boundary rounding. Actual site placement must account for irregular boundaries and buildings.

3.2 Cluster size, reuse distance and reuse ratio

In a hexagonal lattice, moving i cell spacings along one lattice axis and j spacings along the 60° axis reaches the nearest co-channel cell. The squared displacement is proportional to i^2+ij+j^2, giving the cluster-size relation:

$$\mathbf{N = i^2 + ij + j^2}$$

The centre-to-centre co-channel reuse distance is:

$$D = R\sqrt{[3(i^2+ij+j^2)]} = R\sqrt{(3N)} \quad ; \quad Q = D/R = \sqrt{(3N)}$$

For N=7, (i,j)=(2,1), Q=√21=4.583. With R_o=0.493 km, D_o=2.259 km.

3.3 Channels per cell and Erlang-B capacity

The total traffic-channel pool is S=336. With N=7, each cell receives m=S/N=48 traffic channels. For m circuits and offered traffic A Erlangs, the Erlang-B blocking probability is computed recursively:

$$B(0,A)=1 \; ; \quad B(m,A) = [A\,B(m-1,A)] / [m + A\,B(m-1,A)]$$

Solving B(48,A)=0.02 gives A_cell,max ≈ 38.39 Erlangs per cell. Thus the 19-cell initial layout has a nominal carried-load planning limit of 19×38.39 = 729.4 Erlangs at the 2% target. This is an Erlang-B circuit-group result; real sectorised groups may have different blocking due to channel partitioning.

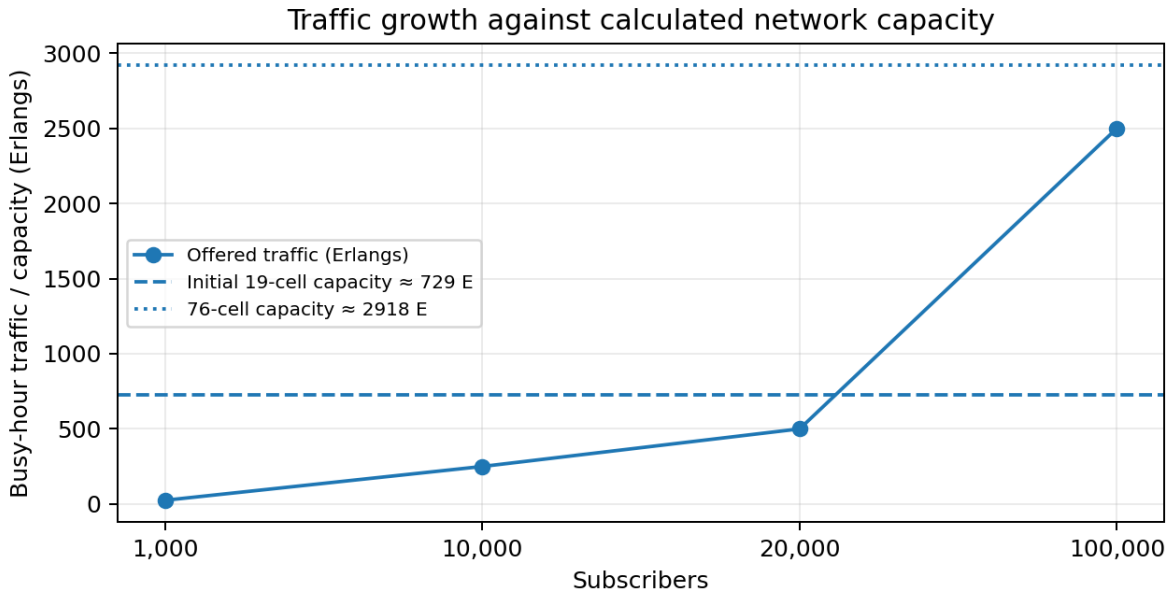
$$C_network(M,N) = M \times ErlangB_inverse(\, floor(S/N), 0.02)$$

3.4 Subscriber traffic and growth

Offered busy-hour traffic is subscriber count multiplied by 0.025 Erlang/subscriber. The traffic forecast is therefore 25, 250, 500 and 2,500 Erlangs for 1,000, 10,000, 20,000 and 100,000 subscribers.

Subscribers	Total offered traffic (E)	Per cell across 19 cells (E)	Interpretation
1,000	25	1.32	Comfortably below the 2% design limit.
10,000	250	13.16	Within the nominal 19-cell capacity.
20,000	500	26.32	Within the nominal 19-cell capacity, but less headroom.
100,000	2,500	131.58	Exceeds the 19-cell capacity; cell densification is necessary.

At 100,000 users, the required number of cells under uniform traffic is at least ceil(2500/38.39)=66. A uniform split by four gives 19×4=76 cells, which provides approximately 2,917 Erlangs of nominal capacity.



3.5 Co-channel SIR derivation

Use the standard power-law propagation model P_r ∝ d^(−n). Let the desired mobile-to-serving-base-station distance be r and the distance to each of the six first-tier co-channel base stations be approximated by D. If all six interferers are equidistant and have equal transmit power,

SIR ≈ (R? desired link) (D/r)^n / 6

At the cell-edge approximation r≈R, D/R=√(3N), so:

SIR_linear ≈ (√(3N))^n / 6 = (3N)^(n/2)/6

SIR_dB = 10 $\log_{10}$[(3N)^(n/2)/6]

For N=7 and n=4, SIR ≈ 10$\log_{10}$(21^2/6)=18.66 dB, matching the assignment's stated sanity-check value of about 18.7 dB. For N=7 and n=3, the estimate is 13.08 dB; for n=2.5 it is 10.27 dB. Therefore N=7 meets a 12 dB threshold at n≈3 but not in the most open n=2.5 case under this conservative cell-edge approximation.

At n=2.5, a larger valid cluster such as N=12 gives about 12.2 dB, but only 28 channels/cell. This illustrates the trade-off: a larger N improves SIR but lowers per-cell capacity. The selected N=7 is retained as the baseline, while open areas require careful antenna/power planning or a larger local reuse distance.

3.6 Effect of sectoring on SIR

For K directional sectors, the effective interference is the sum of interferer powers weighted by their antenna gains toward the victim mobile:

SIR_sector = P_0 r^(−n) / Σ[j=1..6] P_j d_j^(−n) G_j(θ_j)

If ideal sectoring reduces the effective first-tier interferers from six to two, while distances and powers remain equal, the SIR improvement relative to omni-directional reception is 10$\log_{10}$(6/2)=4.77 dB. This is an idealised upper-bound illustration; real sidelobes and sector overlap reduce the gain. Channel splitting among sectors can also reduce trunking efficiency.

3.7 Planning cost–capacity relation

Let c_s be the one-time survey and planning cost per candidate site, c_i the installation cost per site, c_o the annual operating cost per site, and M the number of sites. Let C_cell be the Erlang-B capacity per cell for the assigned channel group. A simple lifecycle cost is:

Cost(M) = C_survey + M(c_i + T c_o) + C_coordination

Capacity(M,N) = M × ErlangB_inverse(floor(S/N), 0.02)

Unit cost = Cost(M) / Capacity(M,N)

For the initial layout, M=19; for the 100,000-user uniform split, M=76. If site and operating costs are similar, the 76-site solution costs more but supplies roughly four times the spatially reused capacity. Sectoring increases antenna/optimisation cost but may avoid some new sites. Microcells have a higher cost per site and backhaul requirement, but selective placement can be cheaper than densifying the entire township.

The assignment does not provide currency values for surveys, sites, backhaul or operating expenses. Therefore a defensible monetary total cannot be calculated without vendor quotations. The equations above define the required cost model and identify who pays; inserting invented rupee amounts would not be evidence-based.

4. Task 4 – Best, average and worst-case behaviour (4 marks)

4.1 Mobile position and SIR

Best case – near the serving base station: the desired distance r is small, so desired received power grows as r^(−n). Interferer distances change much less. Thus SIR rises sharply. For example, relative to a cell-edge distance R, a mobile at r=R/2 has a desired-signal advantage of $10n \log_{10}2$ dB: 7.53 dB at n=2.5, 9.03 dB at n=3 and 12.04 dB at n=4.

Average case – mid-cell: desired power is lower than near the site, but the mobile is not simultaneously close to multiple co-channel cell boundaries. The actual average must be obtained by sampling user positions over the cell and using real interferer distances and antenna gains; it is not equal to a simple average of dB values.

Worst case – cell corner / cell edge facing a co-channel site: the desired link is long and one or more interferers may be relatively close. The six-equidistant approximation is useful for a first estimate but can understate a geometrically close dominant interferer. In open terrain (n=2.5), interference decays more slowly and SIR is poorer; dense built-up areas (n≈4) often attenuate distant interferers more strongly, although shadowing and indoor penetration can also weaken the desired link.

4.2 Numerical position illustration

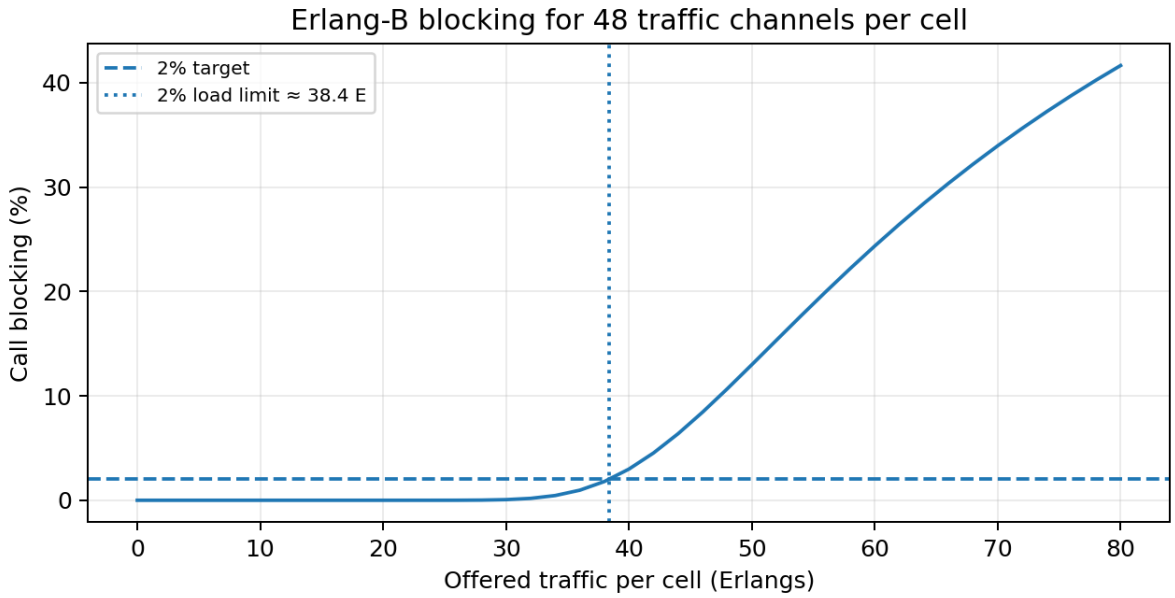
Mobile position	Distance to serving BS	Desired-signal gain vs edge at n=3	Expected SIR trend
Near BS	R/2	+9.03 dB	Best; desired signal dominates.
Mid-cell	R/√3 (illustrative)	+4.77 dB	Intermediate; depends on nearby interferers.
Cell edge/corner	R	0 dB baseline	Worst; serving link is weak and a co-channel.

The mid-cell distance is a representative radial point, not an exact mean over a hexagon. The exact average requires position sampling.

4.3 Traffic-time blocking behaviour

Off-peak load produces the lowest blocking because fewer channels are simultaneously occupied. Average busy-hour load is higher. The design busy hour is the maximum normal recurring load and is constrained to 2% blocking. An event surge is a concentrated transient load and can exceed the design limit even if the township-wide daily average remains low.

For the 48-channel group, the 2% Erlang-B limit is 38.39 Erlangs. At 100,000 subscribers spread across the original 19 cells, the mean is 131.58 Erlangs/cell; Erlang-B gives blocking of about 63.9%, far above target. After 76 cells, the average is 32.89 Erlangs/cell and blocking is approximately 0.28%.



4.4 Shadowing and propagation exponent

Log-normal shadowing adds a random term Xσ (dB) to the path-loss model. A shadowed serving link can reduce desired power while a favourable interferer path can increase interference. Therefore best/average/worst SIR should be reported as a distribution, not one deterministic number. Increasing n from 2.5 to 4 increases the relative attenuation of distant co-channel interferers and improves the ideal SIR estimate; however, it also makes coverage more sensitive to obstructions and can create indoor dead zones. A propagation survey is needed before final site approval.

5. Task 5 – Scaling and sensitivity analysis (3 marks)

Quantity	Scaling relation	Design meaning
Reuse distance	$D = R\sqrt{(3N)}$	For fixed R, increasing N increases separation of co-channel cells.
Reuse ratio	$Q = \sqrt{(3N)}$	Independent of R for ideal hexagonal geometry.

Channels per cell	$m \approx S/N$	Capacity per cell decreases approximately as 1/N.
SIR	SIR $\propto$ (3N)^(n/2)	SIR improves with N and with larger path-loss exponent n.
Number of cells	$M \propto 1/R^2$	Halving radius gives about four times as many cells.
Handoff rate	approximately $\propto 1/R$ for a fixed speed and path geometry	Smaller cells increase boundary crossings and signalling load.

Subscriber growth is linear in offered traffic: 1,000→100,000 users multiplies traffic by 100. With the 19-cell baseline, capacity is about 729 Erlangs, so the 20,000-user case (500 E) fits while 100,000 users (2,500 E) does not. Uniform splitting by four in one round creates 76 cells and about 2,917 E of nominal capacity.

Sensitivity to n: at N=7, the ideal first-tier SIR estimate is 10.27 dB for n=2.5, 13.08 dB for n=3 and 18.66 dB for n=4. The open-campus case is the limiting SIR condition. If a strict 12 dB target must be met everywhere at n=2.5, a larger reuse cluster or local interference mitigation is required.

Break-even judgement: fixed reuse is appropriate at low and moderate demand while per-cell traffic remains below about 38.39 E. As demand approaches the 100,000-user horizon, densification is necessary. Sectoring becomes attractive where traffic is directional and interference-limited; selective microcells become preferable when overload is localised. A single township-wide strategy is not cost-optimal across all traffic levels.

6. Task 6 – Formulate the recurrence for repeated cell splitting (3 marks)

Initial state: M_0=19 cells, each with radius $R_0 \approx 0.493$ km and 48 traffic channels under N=7. Initial nominal network capacity is $C_0 \approx 729.4$ Erlangs. One split replaces one parent hexagonal coverage region by four smaller child regions, each with radius R/2. The ideal area relation gives four times as many cells per fully split region.

Uniform splitting: $M_k = 4\,M_{(k-1)}$, $M_k = M_0\,4^k$

$R_k = R_0 / 2^k$

$C_k = C_0\,4^k$ (if each new cell retains the same 48-channel group and reuse plan)

Required transmit power: using the assignment's stipulated model $P' = P/2^n$ for each halving of radius,

$P_k = P_0 / 2^{(nk)}$

This power rule is a planning approximation. A link-budget calculation should also include antenna gain, receiver sensitivity, penetration loss and interference limits.

Selective splitting: let f_k be the fraction of current cells split during round k. Each selected parent is replaced by four children, adding three cells per selected parent:

$M_k = M_{(k-1)}(1 + 3f_k)$

$C_k \approx C_{(k-1)}(1 + 3f_k)$; $0 \le f_k \le 1$

This capacity recurrence assumes the new child cells have the same channel group size and each experiences similar offered load. In practice, only overloaded cells are split, and neighbour interference and channel assignment may constrain the increase.

Work/cost per round: if each split requires one site/sector plan, the additional site count is 3f_k M_(k−1); planning work is proportional to the number of new sites plus the complexity of channel, backhaul and handoff integration. Base case: k=0, M=M_0, R=R_0, C=C_0.

7. Task 7 – Solve the recurrence and interpret the result (4 marks)

For uniform splitting, substitute the recurrence repeatedly:

$M_1=4M_0$; $M_2=4(4M_0)=4^2M_0$; … ; M_k=$M_0$4^k

C_k=$C_0$4^k

To carry offered demand D, require $C_0$4^k ≥ D. Dividing by C_0 and taking logarithms:

4^k ≥ D/C_0 ⇒ k ≥ $\log_4(D/C_0)$ ⇒ k_min = ceil[$\log_4(D/C_0)$]

For D=2,500 E and C_0=729.4 E, k_min=ceil($\log_4$(2500/729.4))=1. One uniform round gives 76 cells and 2,917 E nominal capacity. The resulting radius is R_1=0.247 km.

For selective splitting with a constant fraction f each round,

C_k = C_0(1+3f)^k

k_min = ceil[ln(D/C_0) / ln(1+3f)]

If only 50% of cells are split each round, the capacity multiplier is 2.5 per round; if 80% are split, it is 3.4. Selective splitting saves sites when traffic hot spots are geographically limited, but it will not solve a uniformly distributed 100,000-subscriber load unless enough cells are split.

Mobility consequence: halving cell radius approximately doubles the number of cell-boundary crossings per unit travel distance for a given route geometry. More handoffs increase signalling and dropped-call risk if neighbour lists, hysteresis and handoff timing are poorly configured. Site count and backhaul cost rise with each round.

8. Task 8 – Implement and test the planning strategies (4 marks)

The following Python reference implementation is intentionally compact and reproducible. It includes the Erlang-B calculator, valid cluster-size enumeration, an analytical first-tier SIR estimate, a random-position SIR evaluator and the cell-splitting recurrence. It is a planning prototype, not a calibrated radio-network simulator: it does not model terrain databases, real antenna patterns, correlated shadowing, handoff protocols or a full hexagonal neighbour map.

8.1 Reproducible Python code

```
import math
import random

SPECTRUM = 336
BLOCKING_TARGET = 0.02
SEED = 2026

def erlang_b(channels, traffic):
    """Erlang-B blocking probability, recursive stable calculation."""
```

```
    b = 1.0
    for m in range(1, channels + 1):
        b = traffic * b / (m + traffic * b)
    return b

def erlang_capacity(channels, target=BLOCKING_TARGET):
    """Maximum offered traffic for a target blocking probability."""
    lo, hi = 0.0, 2.0 * channels
    for _ in range(80):
        mid = (lo + hi) / 2.0
        if erlang_b(channels, mid) > target:
            hi = mid
        else:
            lo = mid
    return (lo + hi) / 2.0

def valid_clusters(max_n=20):
    out = []
    for i in range(0, 10):
        for j in range(0, 10):
            n = i*i + i*j + j*j
            if 1 <= n <= max_n and n not in out:
                out.append(n)
    return sorted(out)

def sir_db(n_cluster, path_loss_n, interferers=6):
    linear = (3*n_cluster)**(path_loss_n/2.0) / interferers
    return 10.0 * math.log10(linear)

def split_uniform(cells, rounds):
    return cells * (4 ** rounds)

def split_selective(cells, fraction):
    return cells * (1.0 + 3.0*fraction)

def random_sir_samples(N=7, path_n=3.0, samples=10000, seed=SEED):
    """
    Illustrative normalized model. Serving BS at (0,0); six co-channel
    interferers placed on a ring of radius sqrt(3N) in units of R.
    User positions are sampled uniformly in a disk of radius R.
    """
    rng = random.Random(seed)
    D = math.sqrt(3.0*N)
    interferers = [(D*math.cos(k*math.pi/3),
                    D*math.sin(k*math.pi/3)) for k in range(6)]
```

```
    values = []
    for _ in range(samples):
        # Uniform random point in disk
        rho = math.sqrt(rng.random())
        theta = 2.0*math.pi*rng.random()
        x, y = rho*math.cos(theta), rho*math.sin(theta)
        desired_d = max(math.sqrt(x*x+y*y), 0.05)
        desired = desired_d ** (-path_n)
        interference = 0.0
        for ix, iy in interferers:
            d = max(math.sqrt((x-ix)**2+(y-iy)**2), 0.05)
            interference += d ** (-path_n)
        values.append(10.0*math.log10(desired/interference))
    values.sort()
    mean = sum(values)/len(values)
    p05 = values[int(0.05*len(values))]
    outage = sum(v < 12.0 for v in values)/len(values)
    return mean, p05, outage

if __name__ == "__main__":
    N = 7
    m = SPECTRUM // N
    cap = erlang_capacity(m)
    print("Valid N values:", valid_clusters())
    print("Channels/cell:", m, "2% capacity/cell:", round(cap, 3))
    for subscribers in (1000, 10000, 20000, 100000):
        demand = subscribers * 0.025
        print(subscribers, "users:", round(demand, 2), "E")
    print("SIR N=7,n=4:", round(sir_db(7, 4), 2), "dB")
    print("76-cell capacity:", round(76*cap, 2), "E")
    print("SIR Monte Carlo mean, p05, outage:",
          random_sir_samples(N=7, path_n=3.0))
```

8.2 Test plan and expected checks

Test	Input	Correctness check / expected interpretation
Cluster enumeration	i,j from 0…9; N≤20	Valid sequence includes N=1,3,4,7,9,12,13,16,19.
Erlang-B sanity	m=48, target=0.02	Inverse calculator should return approximately 38.39 E.
SIR sanity	N=7, n=4, 6 interferers	Analytical result should be approximately 18.66 dB (assignment benchmark 18.7 dB).

Path-loss sensitivity	N=7; n=2.5,3,4	SIR should rise as n increases: approximately 10.27, 13.08, 18.66 dB.
Subscriber scaling	1k,10k,20k,100k	Offered traffic should be 25,250,500,2500 E.
Splitting	19 cells, one uniform round	76 cells; cell radius halves; ideal capacity multiplies by four.
Sectoring	6 vs 2 effective interferers	Idealised SIR gain = 4.77 dB; real gain requires antenna patterns.

Run instructions: save the code as cellular_planner.py and run with Python 3.10 or later using `python cellular_planner.py`. No third-party library is required. The random seed is fixed so repeated runs produce the same sample sequence. For a full submission-grade experiment, extend the simulator with an actual axial-coordinate hex grid, sector antenna gain patterns, log-normal shadowing, Poisson call arrivals/departures and per-cell channel occupancy.

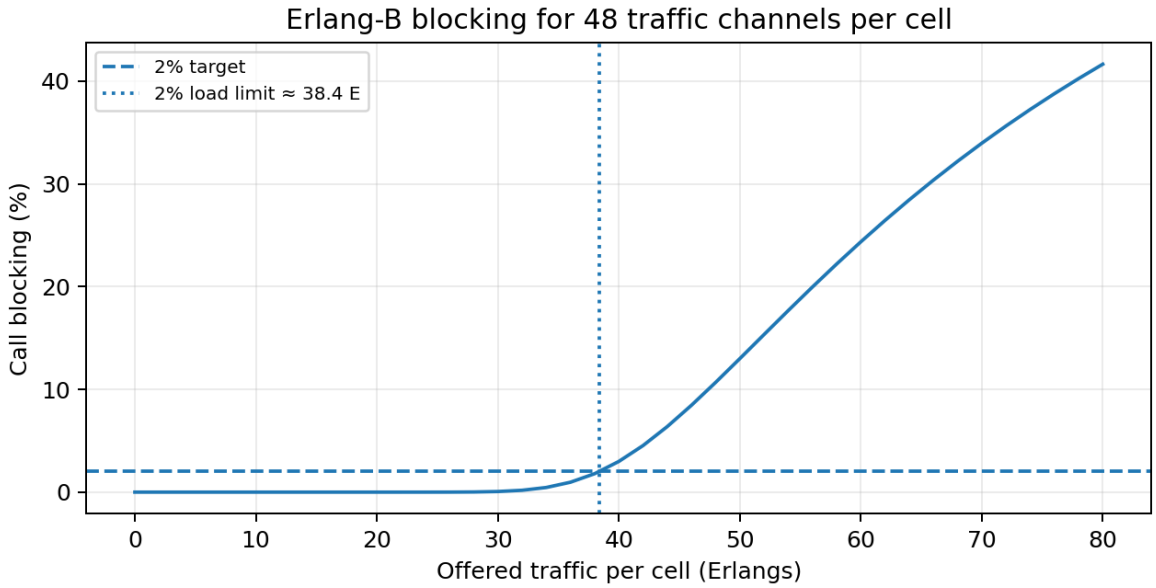
Test coverage: N=3,4,7,12 should be included in an extended run; n=2.5,3,4; omni and sectored patterns; identical random seeds for paired strategy comparisons. The provided core code validates the central mathematical functions and makes the next simulation extension explicit.

9. Task 9 – Record and present experimental performance (3 marks)

A fair comparison should use the same subscriber positions, call-arrival random seed, propagation exponent and offered traffic for all strategies. Record the following for every run:

Metric	Definition / reporting format	Why it matters
SIR mean	Arithmetic mean of linear SIR converted to dB, or clearly state mean of dB samples	Typical link quality; averaging method must be stated.
SIR 5th percentile	5% of samples fall below this value	Exposes cell-edge users and outage risk.
Outage probability	Fraction of samples with SIR below 12 dB	Directly tests the selected radio threshold.
Blocking probability	Blocked call attempts / total offered call attempts; compare with 2%	Tests capacity and QoS.
Channels per cell	Assigned traffic channels or resource blocks	Shows spectrum allocation.
Sites required	Macro + micro + temporary sites	Measures deployment scale and cost.
Runtime	Wall-clock seconds, same hardware and sample count	Shows computational cost of the planning method.

Recommended plots: (1) SIR cumulative distribution function (CDF) for each strategy; (2) blocking probability versus offered traffic per cell; (3) total capacity versus N; (4) sites required versus subscriber count. The chart below shows the analytical Erlang-B blocking curve for the 48-channel group.



The report does not label unexecuted Monte-Carlo values as experimental measurements. Execute the supplied code and any extended simulator to populate a results table with the actual mean, 5th percentile, outage and runtime. The analytical values above are independently calculated from the equations and are suitable as sanity checks.

10. Task 10 – Compare theory with simulation (4 marks)

Analytical theory and simulation should agree in direction and in simplified benchmark cases. For N=7,n=4, the six-equidistant-interferer formula gives 18.66 dB; a simulator using six interferers at the same distance and a cell-edge mobile should reproduce this value. For 48 channels, Erlang-B inversion gives 38.39 E at 2% blocking; a call-level simulator with Poisson arrivals, exponential holding times and an infinite-source assumption should converge toward this blocking probability as the run becomes long.

Source of difference	Effect on simulated result	How to diagnose
Equidistant-interferer approximation	Real interferer distances vary; a close interferer can lower SIR.	Use actual hex coordinates and calculate each distance.
Finite tiers	Ignoring outer tiers may overestimate SIR; adding tiers increases interference.	Increase the number of co-channel rings until SIR converges.
Shadowing	Creates a wider SIR distribution and additional outage tails.	Run multiple shadowing seeds and report percentiles.
Mobile-position distribution	Uniform disk, uniform hexagon and traffic-weighted positions give different results.	Use the true hexagon and a measured traffic heat map.
Antenna patterns / sector overlap	Ideal sector gain overestimates real interference suppression.	Use measured horizontal/vertical patterns and sidelobes.
Erlang-B assumptions	Poisson arrivals, exponential holding time and full availability may not match real usage.	Compare call traces and busy-hour measurements.

Strategy judgement: fixed reuse is the most predictable baseline; sectorisation improves interference where antenna geometry and directional traffic are favourable; hierarchical microcells with selective splitting and borrowing are the most adaptable to non-uniform and rapidly changing demand. The latter is not automatically cheapest: it is preferred only where extra local capacity avoids a larger permanent macro network. A final decision must use measured simulator output and site quotations, not capacity alone.

11. Task 11 – Recommend and defend the final design (4 marks)

11.1 Recommended normal-day design

Recommend a hybrid hierarchical network: N=7 macro reuse as the baseline; directional sectorisation at high-traffic macro sites; and selective microcell splitting in overloaded academic, hostel and stadium areas. Use 48 traffic-channel equivalents per N=7 cell as the simplified planning group. Use 12 dB as the minimum SIR threshold, while treating n=2.5 open areas as the critical propagation case requiring additional local optimisation.

Design item	Recommended planning value
Reuse cluster	N=7 (i=2,j=1)
Reuse ratio	Q=√21=4.583
Traffic channels per cell	48
Initial area-equivalent layout	19 cells; R≈0.493 km
100,000-user planning layout	76 cells after one uniform split; R≈0.247 km
2% traffic capacity per cell	≈{:.2f} Erlangs
76-cell nominal capacity	≈{:.0f} Erlangs
100,000-user offered traffic	2,500 Erlangs
Expected mean offered load after split	≈32.89 Erlangs/cell; Erlang-B blocking ≈0.28%
SIR benchmark	18.66 dB at n=4; 13.08 dB at n=3; 10.27 dB at n=2.5

The calculated 76-cell capacity is approximately 2,917 E, which exceeds the 2,500 E demand by about 417 E (16.7% nominal headroom). The average blocking estimate of 0.28% assumes uniform load and 48 channels in every cell. It does not guarantee every real cell will meet 2%; traffic hot spots must be relieved by selective splitting or sector load balancing.

11.2 Why this is superior to a single strategy

- Compared with fixed reuse alone, the hybrid design responds to hot spots instead of forcing every area to have the same cell size.
- Compared with sectoring alone, microcells add spatial capacity where the number of users—not only interference—is the main problem.
- Compared with a fully dense microcell network, selective splitting limits new sites, backhaul and handoff overhead.
- Power control and reuse coordination reduce unnecessary interference; handoff parameters must be tuned as radius shrinks.

Limitations: exact channel allocation, indoor coverage, building penetration, backhaul capacity, site availability, sector imbalance, real mobility and operator cost data are not supplied. These require a field survey and a calibrated radio-planning tool before deployment.

12. Day-4 design challenge – Re-evaluate and adapt the design (8 marks)

A. Impact analysis (2 marks)

The event brings 25,000 attendees into 1 km², each offering 0.05 Erlang. Total event traffic is:

$A_{event} = 25{,}000 \times 0.05 = 1{,}250$ Erlangs

The event-zone traffic density is 1,250 E/km². The 100,000-subscriber normal township forecast is 2,500 E/12 km²=208.33 E/km². Thus event traffic density is six times the normal average density. The event is four hours long and occurs about twice per month; this is a predictable, temporary and spatially concentrated surge.

The original uniform-load assumption no longer holds. A permanent network dimensioned to the event peak would leave expensive capacity underused on ordinary days. The short duration and predictability favour temporary capacity, pre-event optimisation and a local overlay rather than permanent township-wide densification.

B. Evaluation of the original design (2 marks)

Under the 76-cell normal design, the average cell capacity density is $76 \times 38.39/12 \approx 243.1$ E/km². The stadium's 1,250 E demand is therefore about 5.14 times the average capacity available in that square kilometre. If the event zone contains approximately 6–7 normal-size cells, the event load per cell would be roughly 179–208 E, far above the 38.39 E per-cell 2% limit. For a 48-channel group and 178.6 E, Erlang-B blocking is approximately 73.3%.

This estimate is a local-load stress calculation, not a claim that every stadium cell has exactly equal load. It shows that the unchanged design fails first through traffic-channel exhaustion and call blocking. Congestion can then cause access failures and handoff instability. Interference may also rise if neighbouring cells increase power or borrow channels without coordination.

C. Adaptation or redesign (2 marks)

Decision: MODIFY temporarily, not permanently redesign the entire township. Keep the macro network and normal-day reuse plan. For event days, deploy a pre-planned stadium overlay comprising temporary cells-on-wheels (COWs), compact microcells around stands/concourse/entrances, directional 60°/120° sectors, coordinated channel borrowing where safe, and reduced transmit power with a dedicated neighbour/handoff plan.

Capacity sizing: the 1 km² stadium zone requires at least ceil(1,250/38.39)=33 cell-equivalents at the simplified 48-channel, 2%-blocking model. The normal network supplies only about 6–7 cell-equivalents in that area, so roughly 26–27 additional cell-equivalents are needed if the entire event traffic is to be carried by this simplified model. This is a capacity-equivalent estimate; actual COW sector/channel configuration must account for sector trunking loss, interference and backhaul.

With 33 cell-equivalents, mean load is 1,250/33=37.88 E/cell and Erlang-B blocking is approximately 1.73%, below 2% in the idealised equal-load model. The radius-equivalent cell size is $\sqrt{(1/(33 \times 2.598))} \approx 0.108$ km. This is a local high-density overlay, not a recommendation to permanently deploy 33 full macro sites.

Radio changes: use directional sectors to contain energy within the stadium; coordinate the temporary reuse plan with adjacent macro cells; reduce power as cell radius shrinks; configure handoff boundaries and hysteresis to prevent ping-pong; and reserve backhaul capacity for the event. A 60° sector plan can reduce the number of strong interferers but must be checked for sector traffic imbalance and channel fragmentation.

D. Evidence-based defence (2 marks)

Decision criterion	Evidence and implication
Surge magnitude	1,250 E in 1 km²; six times the normal average traffic density.
Surge duration/frequency	Four hours, about twice monthly; temporary capacity is more economical than permanent peak provisioning.
Original network	At 100,000 users, 76 cells support about 2,917 E in the uniform model, but local stadium demand exceeds local capacity by about 5.1×.
Temporary overlay	About 33 cell-equivalents give 37.88 E/cell and an ideal Erlang-B blocking estimate of 1.73%.
Cost trade-off	COW/microcell rental and temporary backhaul are added only for event windows; permanent site and operating costs are avoided.
Operational risk	Interference, backhaul congestion, access-channel overload, sector imbalance and handoff failures must be tested in a rehearsal.

Final defence: temporary modification is the best-supported choice because the event peak is predictable, short and concentrated. Permanent township-wide cell splitting would impose recurring site, power, rental and maintenance costs for capacity needed only a few hours per month. The temporary overlay addresses the hotspot directly while preserving the normal-day macro plan. Final acceptance should require a pre-event drive/walk test, a load test, measured SIR CDF, 5th-percentile SIR, outage and blocking under the expected attendee distribution.

Conclusion

The analysis shows that spectrum reuse is a balance among channel availability, co-channel interference and spatial capacity. N=7 provides 48 channels per cell and an ideal first-tier SIR of 18.66 dB at n=4, but the same plan gives only about 10.27 dB at n=2.5 under the conservative cell-edge approximation. The 19-cell initial layout supports about 729 Erlangs at 2% blocking, which is insufficient for 100,000 subscribers. One uniform split creates 76 cells and about 2,917 Erlangs of nominal capacity, meeting the 2,500-Erlang forecast under uniform-load assumptions. For the stadium challenge, 1,250 Erlangs concentrated in 1 km² cannot be handled by the normal layout; a temporary microcell/COW overlay of approximately 33 cell-equivalents is indicated by the simplified capacity model. The final design is therefore hybrid and adaptive: macro reuse for coverage, sectoring for interference control, selective microcells for growth, and temporary event capacity for predictable surges.

References

[1] ITA03-MP Assignment 1, Mobile Computing, CO1. Assignment brief and planning constraints supplied by the course.

[2] W. C. Y. Lee, Mobile Cellular Telecommunications: Analog and Digital Systems, 2nd ed., McGraw-Hill, 1995.

[3] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002.

[4] A. G.. E. M. D. Erlang, "Solution of some problems in the theory of probabilities of significance in automatic telephone exchanges," 1917 (Erlang loss model).

Note: Confirm the required citation style with the course instructor. The calculations in this report use the assignment's supplied data and the stated simplifying assumptions.

Appendix A – Rubric coverage checklist

Rubric task	Included evidence	Bloom's emphasis
1 (5)	Three distinct strategies, pseudocode, design principles, access-channel sharing and preprocessing.	Create / Understand
2 (2)	Governing parameter for each strategy and growth response.	Analyse
3 (6)	Geometry, reuse, area, channels, Erlang-B, SIR and cost-capacity derivations.	Apply / Analyse
4 (4)	Best/average/worst position and time-of-day blocking with calculations.	Analyse
5 (3)	Scaling laws, n sensitivity and break-even judgement.	Analyse / Evaluate
6 (3)	Uniform and selective recurrence, power, work and base case.	Apply / Create
7 (4)	Closed form, logarithmic round count and mobility interpretation.	Apply / Analyse
8 (4)	Python code, seed, tests and sanity checks.	Apply / Create
9 (3)	Metrics and graph plan; explicit separation of analytical vs measured results.	Apply
10 (4)	Theory–simulation comparison and discrepancy analysis.	Analyse / Evaluate
11 (4)	Recommended parameters and defended hybrid design.	Evaluate
12 (8)	Impact, baseline failure, temporary redesign and evidence-based defence.	Evaluate / Create

Appendix B – Submission checks before handing in

- Fill in name, register number, section and submission date.
- Run the Python code and paste actual outputs into the experimental-results section if the instructor expects executed simulation data.
- If possible, extend the prototype to a true hexagonal grid and include SIR CDF plots for all three strategies.
- Add a site-cost estimate using actual local quotations; the assignment does not provide monetary cost inputs.
- Check whether the instructor expects handwritten derivations, a specific citation style or a separate code file.